

SDS8102: Advanced Mathematical Statistics

Lecture 5: Foundations of Hypothesis Testing

Saptarshi Roy

Fall 2026

Department of Statistics and Data Science
MBZUAI

Lecture 5: Hypothesis Testing

Estimation asks:

What value of the parameter is plausible given the data?

Hypothesis testing asks a different question:

Is the observed data compatible with a particular claim about the parameter?

Examples include:

- Does a new treatment have an effect?
- Is a coin fair?
- Is a regression coefficient zero?
- Do two populations have the same mean?

Today we develop a mathematical framework for answering such questions.

Learning Objectives

By the end of this lecture, you should be able to

1. Formulate null and alternative hypotheses.
2. Define a statistical test and rejection region.
3. Distinguish Type I and Type II errors.
4. Define and interpret the power function.
5. Distinguish the size and level of a test.
6. Construct simple tests in the Gaussian model.

A Running Example: Does a Treatment Work?

Suppose $X_1, \dots, X_n$ measure the improvement of patients after receiving a treatment.

Assume

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known.

The treatment has no average effect if

$$\mu = 0.$$

We therefore want to distinguish between

$H_0 : \mu = 0$	versus	$H_1 : \mu > 0.$
-----------------	--------	------------------

How should the observed data determine which hypothesis we favor?

Null and Alternative Hypotheses

Let

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

be a statistical model.

Partition the parameter space into two sets:

$$\Theta_0 \quad \text{and} \quad \Theta_1.$$

We wish to test

$H_0 : \theta \in \Theta_0$	versus	$H_1 : \theta \in \Theta_1.$
-----------------------------	--------	------------------------------

H_0 is called the **null hypothesis** and H_1 the **alternative hypothesis**.

Simple and Composite Hypotheses

A hypothesis is **simple** if it specifies a single distribution.

For example,

$$H_0 : \mu = 0$$

is simple in the model $N(\mu, 1)$.

A hypothesis is **composite** if it contains multiple distributions.

For example,

$$H_1 : \mu > 0$$

is composite because it contains every distribution

$$N(\mu, 1), \quad \mu > 0.$$

This distinction will become important when we search for **optimal tests**.

What Is a Statistical Test?

A statistical test is a rule that maps the observed data into a decision.

Definition: Test Function

A nonrandomized test is a function

$$\phi : \mathcal{X} \rightarrow \{0, 1\},$$

where

$$\phi(X) = \begin{cases} 1, & \text{reject } H_0, \\ 0, & \text{do not reject } H_0. \end{cases}$$

Thus hypothesis testing is itself a statistical decision problem.

The Rejection Region

Every nonrandomized test can equivalently be described by a **rejection region**

$$R = \{x : \phi(x) = 1\}.$$

The decision rule is then

$$X \in R \implies \text{reject } H_0.$$

For our treatment example, a natural rule might be

$$R = \{\bar{X} > c\}.$$

Large positive values of $\bar{X}$ provide evidence against

$$H_0 : \mu = 0$$

in favor of $H_1 : \mu > 0$.

But how should we choose c ?

Two Ways a Test Can Be Wrong

There are two possible errors.

	Do not reject H_0	Reject H_0
H_0 true	Correct	Type I error
H_1 true	Type II error	Correct

A **Type I error** occurs when we reject a true null hypothesis.

A **Type II error** occurs when we fail to reject a false null hypothesis.

These two errors compete with one another.

Making rejection easier reduces Type II error but increases the chance of Type I error.

The Fundamental Trade-Off

Return to the test

$$\text{reject } H_0 \quad \text{if } \bar{X} > c.$$

If we decrease c , rejection becomes easier:

$$\text{smaller } c \Rightarrow \begin{cases} \text{more Type I errors,} \\ \text{fewer Type II errors.} \end{cases}$$

If we increase c , rejection becomes harder:

$$\text{larger } c \Rightarrow \begin{cases} \text{fewer Type I errors,} \\ \text{more Type II errors.} \end{cases}$$

We therefore need a principled way to control one error while optimizing the other.

The Power Function

For a test ϕ , define its **power function** by

$$\beta_{\phi}(\theta) = \mathbb{P}_{\theta}(\phi(X) = 1) = \mathbb{E}_{\theta}[\phi(X)].$$

Thus $\beta_{\phi}(\theta)$ is the probability that the test rejects H_0 when the true parameter is θ .

Its interpretation depends on where θ lies:

$\theta \in \Theta_0$: probability of Type I error,

$\theta \in \Theta_1$: probability of correctly rejecting H_0 .

Power in the Gaussian Example

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

and reject $H_0 : \mu = 0$ when

$$\bar{X} > c.$$

Since

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right),$$

the power function is

$$\begin{aligned}\beta(\mu) &= P_\mu(\bar{X} > c) \\ &= P_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} > \frac{\sqrt{n}(c - \mu)}{\sigma}\right).\end{aligned}$$

Therefore,

$$\boxed{\beta(\mu) = 1 - \Phi\left(\frac{\sqrt{n}(c - \mu)}{\sigma}\right)}.$$

What Should a Good Power Function Look Like?

For

$$H_0 : \mu \leq 0 \quad \text{versus} \quad H_1 : \mu > 0,$$

we want two things.

Under the null,

$$\beta(\mu) \quad \text{should be small.}$$

Under the alternative,

$$\beta(\mu) \quad \text{should be large.}$$

Ideally,

$$\beta(\mu) \approx \begin{cases} 0, & \mu \leq 0, \\ 1, & \mu > 0. \end{cases}$$

Finite samples prevent this ideal separation.

The goal of test construction is therefore to obtain the best possible power while

Size of a Test

For a composite null

$$H_0 : \theta \in \Theta_0,$$

the probability of rejection may differ across values of θ in Θ_0 .

The **size** of a test ϕ is

$$\text{size}(\phi) = \sup_{\theta \in \Theta_0} \beta_{\phi}(\theta).$$

Thus the size is the **largest probability of Type I error** over all distributions satisfying the null hypothesis.

Level of a Test

A test is said to have **level** α if

$$\sup_{\theta \in \Theta_0} P_{\theta}(\text{reject } H_0) \leq \alpha.$$

Equivalently,

$$\text{size}(\phi) \leq \alpha.$$

Therefore:

- **size** is a property of a particular test;
- **level** α is a constraint imposed on the test.

A level-0.05 test need not have size exactly 0.05.

It may have size strictly smaller than 0.05.

Choosing the Critical Value

Return to

$$H_0 : \mu \leq 0 \quad \text{versus} \quad H_1 : \mu > 0.$$

Reject when

$$\bar{X} > c.$$

The rejection probability increases with μ , so under H_0 the largest rejection probability occurs at the boundary

$$\mu = 0.$$

To construct a level- α test, choose c such that

$$P_{\mu=0}(\bar{X} > c) = \alpha.$$

Since

$$\frac{\sqrt{n}\bar{X}}{\sigma} \sim N(0, 1) \implies \boxed{c = z_{1-\alpha} \frac{\sigma}{\sqrt{n}}}.$$

The Level- α Gaussian Test

We therefore reject

$$H_0 : \mu \leq 0$$

whenever

$$\frac{\sqrt{n} \bar{X}}{\sigma} > z_{1-\alpha}.$$

Its power function is

$$\beta(\mu) = 1 - \Phi \left(z_{1-\alpha} - \frac{\sqrt{n} \mu}{\sigma} \right).$$

This formula reveals how power depends on

- the significance level α ,
- the effect size μ ,
- the sample size n ,
- the noise level σ .

But Why This Test Statistic?

For the Gaussian problem we chose

$$T(X) = \frac{\sqrt{n}\bar{X}}{\sigma}.$$

This choice seems natural because $\bar{X}$ contains information about μ .

But in a general statistical model, there may be many possible test statistics and many possible rejection regions.

Among all level- α tests, which one has the largest power?

This is an **optimization problem over statistical tests**.

The likelihood provides a remarkably powerful answer.

Simple versus Simple Hypotheses

Consider testing two completely specified distributions:

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1.$$

Let their densities be

$$p_{\theta_0}(x) \quad \text{and} \quad p_{\theta_1}(x).$$

For an observed value $X = x$, a natural question is:

Which hypothesis makes the observed data more likely?

This suggests comparing the two likelihoods.

The Likelihood Ratio

Define the likelihood ratio

$$\Lambda(x) = \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)}.$$

Its interpretation is immediate:

$$\Lambda(x) > 1$$

means that the observed data are more likely under H_1 than under H_0 .

More generally, large values of

$$\frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}$$

provide evidence in favor of H_1 .

This suggests rejection regions of the form

$$R = \{x : \Lambda(x) > c\}.$$

The Optimization Problem

For a test ϕ , under the simple null the Type I error is

$$\mathbb{E}_{\theta_0}[\phi(X)].$$

Its power against θ_1 is

$$\mathbb{E}_{\theta_1}[\phi(X)].$$

We therefore want to solve

$$\begin{array}{ll} \text{maximize} & \mathbb{E}_{\theta_1}[\phi(X)] \\ \text{subject to} & \mathbb{E}_{\theta_0}[\phi(X)] \leq \alpha. \end{array}$$

In words:

Among all tests with Type I error at most α , find the test with the largest probability of detecting the alternative.

Definition: Most Powerful Test

A level- α test ϕ^* is **most powerful** against θ_1 if

$$\mathbb{E}_{\theta_1}[\phi^*(X)] \geq \mathbb{E}_{\theta_1}[\phi(X)]$$

for every other level- α test ϕ .

Thus a most powerful test solves an optimization problem over all possible statistical decision rules.

The Neyman–Pearson lemma tells us exactly what this optimal test looks like for simple versus simple hypotheses.

The Neyman–Pearson Lemma

Theorem (Neyman–Pearson Lemma)

Consider

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1.$$

Suppose a level- α test ϕ^ has the form*

$$\phi^*(x) = \begin{cases} 1, & p_{\theta_1}(x) > k p_{\theta_0}(x), \\ 0, & p_{\theta_1}(x) < k p_{\theta_0}(x), \end{cases}$$

with possible randomization when equality holds.

Then ϕ^ is a most powerful level- α test.*

Equivalently, reject H_0 for sufficiently large values of

$$\frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}.$$

Why Should the Likelihood Ratio Be Optimal?

Imagine that we can spend only an α amount of probability under H_0 on the rejection region.

Which observations should we put into that region?

We should choose observations for which

$$\frac{p_{\theta_1}(x)}{p_{\theta_0}(x)}$$

is largest.

Such observations are

- relatively unlikely under H_0 ,
- relatively likely under H_1 .

Thus the likelihood-ratio test spends the available Type I error budget where it produces the greatest gain in power.

Proof of the Neyman–Pearson Lemma

Let ϕ^* be the likelihood-ratio test and let ϕ be any other level- α test.

By construction,

$$(\phi^*(x) - \phi(x))(p_{\theta_1}(x) - kp_{\theta_0}(x)) \geq 0.$$

Integrating,

$$\int (\phi^* - \phi)(p_{\theta_1} - kp_{\theta_0}) dx \geq 0.$$

Therefore,

$$\mathbb{E}_{\theta_1}[\phi^*] - \mathbb{E}_{\theta_1}[\phi] \geq k(\mathbb{E}_{\theta_0}[\phi^*] - \mathbb{E}_{\theta_0}[\phi]).$$

If ϕ^* has size α and ϕ has level α , then

$$\mathbb{E}_{\theta_0}[\phi^*] - \mathbb{E}_{\theta_0}[\phi] \geq 0.$$

Hence

$$\boxed{\mathbb{E}_{\theta_1}[\phi^*] \geq \mathbb{E}_{\theta_1}[\phi].}$$

Thus ϕ^* is most powerful.

Example: Testing a Gaussian Mean

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known.

Consider

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu = \mu_1,$$

where

$$\mu_1 > \mu_0.$$

By the Neyman–Pearson lemma, the most powerful test rejects for large values of

$$\frac{L(\mu_1; X)}{L(\mu_0; X)}.$$

What does this likelihood ratio look like?

The Gaussian Likelihood Ratio

Ignoring constants,

$$L(\mu) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right\}.$$

Hence

$$\log \frac{L(\mu_1)}{L(\mu_0)} = -\frac{1}{2\sigma^2} \left[\sum_i (X_i - \mu_1)^2 - \sum_i (X_i - \mu_0)^2 \right].$$

Simplifying,

$$\log \frac{L(\mu_1)}{L(\mu_0)} = \frac{n(\mu_1 - \mu_0)}{\sigma^2} \bar{X} - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}.$$

Since $\mu_1 > \mu_0$, the likelihood ratio is increasing in $\bar{X}$.

The Most Powerful Gaussian Test

The Neyman–Pearson rejection region therefore has the form

$$\bar{X} > c.$$

Choose c so that the test has size α :

$$P_{\mu_0}(\bar{X} > c) = \alpha.$$

Since

$$\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \sim N(0, 1)$$

under H_0 , we obtain

Reject H_0 if $\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > z_{1-\alpha}.$
--

This is not merely a reasonable test.

It is the **most powerful level- α test** against $\mu = \mu_1$.

Uniformly most powerful test

For every fixed alternative

$$\mu_1 > \mu_0,$$

the Neyman–Pearson lemma gave the same form of rejection region:

$$\bar{X} > c.$$

Moreover, after imposing level α , the critical value c depends on μ_0 , σ , n , and α —but **not on** μ_1 .

Therefore the same test is most powerful against every alternative

$$\mu_1 > \mu_0.$$

This suggests a stronger notion of optimality:

Can one test be most powerful simultaneously against every alternative?

Such a test is called **uniformly most powerful**.

Uniformly Most Powerful Tests

Consider

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta_1.$$

Definition: Uniformly Most Powerful Test

A level- α test ϕ^* is **uniformly most powerful (UMP)** if, for every level- α test ϕ ,

$$\beta_{\phi^*}(\theta) \geq \beta_{\phi}(\theta) \quad \text{for every } \theta \in \Theta_1.$$

Thus ϕ^* is simultaneously most powerful against **every parameter value in the alternative**.

Most Powerful versus Uniformly Most Powerful

A most powerful test solves a simple-versus-simple problem:

$$\theta_0 \text{ versus } \theta_1.$$

A UMP test must solve many such problems simultaneously:

$$\theta_0 \text{ versus } \theta \text{ for every } \theta \in \Theta_1.$$

Therefore,

UMP is a much stronger requirement than most powerful.

Different alternatives may favor different rejection regions.

For this reason, UMP tests do not exist in every testing problem.

Why Did a UMP Test Exist for the Gaussian Mean?

Recall

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

and

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu > \mu_0.$$

For every $\mu_1 > \mu_0$, we found

$$\frac{L(\mu_1; X)}{L(\mu_0; X)}$$

to be an increasing function of $\bar{X}$.

Thus every simple alternative $\mu_1 > \mu_0$ favors the **same ordering of the sample space**:

$\text{larger } \bar{X} \implies \text{stronger evidence for larger } \mu.$

This common ordering is the key.

Monotone Likelihood Ratio

Definition: Monotone Likelihood Ratio

A family $\{P_\theta : \theta \in \Theta\}$ has a **monotone likelihood ratio** in a statistic $T(X)$ if, whenever

$$\theta_1 > \theta_0,$$

the likelihood ratio

$$\frac{p_{\theta_1}(x)}{p_{\theta_0}(x)}$$

is a nondecreasing function of $T(x)$.

In words:

Larger $T(X)$ systematically favors larger values of θ .

Why MLR Is Powerful

Suppose the family has monotone likelihood ratio in $T(X)$.

For any

$$\theta_1 > \theta_0,$$

the Neyman–Pearson lemma says to reject $H_0 : \theta = \theta_0$ for large values of

$$\frac{p_{\theta_1}(X)}{p_{\theta_0}(X)}.$$

But MLR implies that this is equivalent to rejecting for large values of

$$T(X).$$

Crucially, the ordering by $T(X)$ is the same for every $\theta_1 > \theta_0$.

This is what makes uniform optimality possible.

Theorem (Karlin–Rubin)

Suppose $\{P_\theta : \theta \in \Theta\}$ has monotone likelihood ratio in $T(X)$.

For testing

$$H_0 : \theta \leq \theta_0 \quad \text{versus} \quad H_1 : \theta > \theta_0,$$

a test of the form

$$\phi(X) = \mathbf{1}\{T(X) > c\}$$

with c chosen to have level α is UMP among level- α tests.

For a decreasing likelihood ratio, the direction of the rejection region is reversed.

Why the Boundary of the Null Matters

Suppose the rejection region is

$$T(X) > c.$$

Under monotone likelihood ratio, the rejection probability

$$P_{\theta}(T(X) > c)$$

is nondecreasing in θ .

Therefore, under

$$H_0 : \theta \leq \theta_0,$$

the largest Type I error occurs at

$$\theta = \theta_0.$$

Hence c can be chosen by solving

$$P_{\theta_0}(T(X) > c) = \alpha.$$

Example: Gaussian Mean Has MLR

For

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

consider $\mu_1 > \mu_0$.

We previously obtained

$$\log \frac{L(\mu_1)}{L(\mu_0)} = \frac{n(\mu_1 - \mu_0)}{\sigma^2} \bar{X} - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}.$$

Since

$$\frac{n(\mu_1 - \mu_0)}{\sigma^2} > 0,$$

the likelihood ratio is increasing in

$$T(X) = \bar{X}.$$

Therefore the Normal location family has monotone likelihood ratio in $\bar{X}$.

UMP Test for the Gaussian Mean

Consider

$$H_0 : \mu \leq \mu_0 \quad \text{versus} \quad H_1 : \mu > \mu_0.$$

By the monotone likelihood ratio property, the UMP level- α test rejects for large $\bar{X}$.

Choosing the critical value at $\mu = \mu_0$ gives

$$\text{Reject } H_0 \quad \text{if} \quad \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > z_{1-\alpha}.$$

Thus the familiar one-sided z-test is not merely intuitive:

It is UMP among all level- α tests.

Example: Bernoulli Model

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Let

$$T = \sum_{i=1}^n X_i.$$

For $p_1 > p_0$,

$$\frac{L(p_1)}{L(p_0)} = \left(\frac{p_1}{p_0}\right)^T \left(\frac{1-p_1}{1-p_0}\right)^{n-T}.$$

Rearranging,

$$\frac{L(p_1)}{L(p_0)} = \left(\frac{p_1/(1-p_1)}{p_0/(1-p_0)}\right)^T \left(\frac{1-p_1}{1-p_0}\right)^n.$$

Since $p_1 > p_0$, this is increasing in T .

UMP Test for a Binomial Probability

Therefore the Bernoulli family has monotone likelihood ratio in

$$T = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p_0)$$

For testing

$$H_0 : p \leq p_0 \quad \text{versus} \quad H_1 : p > p_0,$$

the UMP level- α test rejects for large values of T :

$$\boxed{T \geq c.}$$

The critical value is chosen so that

$$P_{p_0}(T \geq c) \leq \alpha \quad (c \text{ can be chosen using Binomial quantiles.})$$

A Subtlety: Discrete Test Statistics

Suppose

$$T \sim \text{Binomial}(n, p_0).$$

Because T takes only integer values, there may be no critical value c satisfying

$$P_{p_0}(T \geq c) = \alpha$$

exactly.

We can choose c so that

$$P_{p_0}(T \geq c) \leq \alpha,$$

giving a valid level- α test.

But its size may be strictly smaller than α .

This is one reason randomized tests appear naturally in the general theory.

What About Two-Sided Alternatives?

So far the alternatives have had a preferred direction:

$$H_1 : \theta > \theta_0.$$

Monotone likelihood ratio tells us that large $T(X)$ provides evidence in that direction.

Now consider

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta \neq \theta_0.$$

For $\theta > \theta_0$, we want to reject for **large** $T(X)$.

But for $\theta < \theta_0$, we typically want to reject for **small** $T(X)$.

Thus the most powerful rejection region depends on which side of the null the alternative lies.

A UMP test may therefore fail to exist for two-sided alternatives.

The Difficulty with Two-Sided Alternatives

Consider

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta \neq \theta_0.$$

Suppose the model has monotone likelihood ratio in $T(X)$.

For alternatives

$$\theta > \theta_0,$$

the Neyman–Pearson principle favors rejection for **large** values of $T(X)$.

But for alternatives

$$\theta < \theta_0,$$

it favors rejection for **small** values of $T(X)$.

Thus the two directions compete for the available Type I error.

Example: Two-Sided Gaussian Mean

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known.

Consider

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0.$$

For any fixed $\mu_1 > \mu_0$, the most powerful level- α test rejects for large values of $\bar{X}$.

For any fixed $\mu_1 < \mu_0$, the most powerful level- α test rejects for small values of $\bar{X}$.

The most powerful rejection regions point in opposite directions.

Why a UMP Test Does Not Exist

Suppose a UMP level- α test existed for

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0.$$

It would have to be most powerful against every $\mu_1 > \mu_0$.

By the Neyman–Pearson lemma, this requires spending the Type I error in the **upper tail**.

But it must also be most powerful against every $\mu_1 < \mu_0$, which requires spending the Type I error in the **lower tail**.

These requirements cannot generally be satisfied simultaneously.

There is no UMP level- α test for the usual two-sided Gaussian mean problem.

A Weaker Requirement: Unbiased Tests

Recall that the power function is

$$\beta_{\phi}(\theta) = P_{\theta}(\text{reject } H_0).$$

A natural requirement is that the test should be more likely to reject under the alternative than under the null.

Definition: Unbiased Test

A level- α test is **unbiased** if

$$\beta_{\phi}(\theta) \geq \alpha \quad \text{for every } \theta \in \Theta_1.$$

Thus moving away from the null should not make rejection *less likely* than its nominal level.

Why Unbiasedness Helps

For the two-sided problem

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0,$$

an unbiased level- α test satisfies

$$\beta(\mu_0) = \alpha$$

and

$$\beta(\mu) \geq \alpha \quad \text{for every } \mu \neq \mu_0.$$

Thus the power function has its minimum at the null:

$\mu_0 \text{ is a minimizer of } \beta(\mu).$

For a smooth power function, this suggests the condition

$$\beta'(\mu_0) = 0.$$

Uniformly Most Powerful Unbiased Tests

Once we restrict attention to unbiased tests, we can again ask for optimality.

Definition: UMPU Test

A level- α test ϕ^* is **uniformly most powerful unbiased (UMPU)** if it is unbiased and

$$\beta_{\phi^*}(\theta) \geq \beta_{\phi}(\theta) \quad \text{for every } \theta \in \Theta_1$$

for every other unbiased level- α test ϕ .

The restriction to unbiased tests makes optimal two-sided procedures possible in many important models.

The Two-Sided Gaussian Test

For

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0,$$

define

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}.$$

Under H_0 ,

$$Z \sim N(0, 1).$$

The familiar two-sided level- α test rejects when

$$|Z| > z_{1-\alpha/2}.$$

Equivalently,

$$Z < -z_{1-\alpha/2} \quad \text{or} \quad Z > z_{1-\alpha/2}.$$

The Type I error α is split equally between the two tails.

Power of the Two-Sided Gaussian Test

Under a general value μ ,

$$Z \sim N\left(\frac{\sqrt{n}(\mu - \mu_0)}{\sigma}, 1\right).$$

Let

$$\delta = \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}.$$

The power is

$$\begin{aligned}\beta(\mu) &= P_\mu(|Z| > z_{1-\alpha/2}) \\ &= P_\mu(Z < -z_{1-\alpha/2}) + P_\mu(Z > z_{1-\alpha/2}).\end{aligned}$$

Therefore,

$$\boxed{\beta(\mu) = \Phi(-z_{1-\alpha/2} - \delta) + 1 - \Phi(z_{1-\alpha/2} - \delta).}$$

The Shape of the Power Function

At the null,

$$\beta(\mu_0) = \alpha.$$

As μ moves away from μ_0 in either direction,

$$\beta(\mu) \longrightarrow 1.$$

Moreover, the power function is symmetric around μ_0 :

$$\boxed{\beta(\mu_0 + h) = \beta(\mu_0 - h).}$$

Thus

$$\beta'(\mu_0) = 0,$$

and the test is unbiased.

In the Normal model with known variance, this familiar two-sided test is in fact **UMPU**.

A Hierarchy of Optimal Tests

We have now encountered three notions of optimality.

Setting	Optimality	Main tool
Simple vs. simple	Most powerful	Neyman–Pearson
One-sided composite	UMP	Monotone likelihood ratio
Two-sided composite	Often no UMP	UMP unbiased

The key distinction is:

One-sided alternatives have a direction; two-sided alternatives compete across directions.

This is why two-sided optimal testing requires additional structure or restrictions.